

Aluminium and lead: molecular mechanisms of brain toxicity.

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The fact that aluminium (Al) and lead (Pb) are both toxic metals to living organisms, including human beings, was discovered a long time ago. Even when Al and Pb can reach and accumulate in almost every organ in the human body, the central nervous system is a particular target of the deleterious effects of both metals. Select human population can be at risk of Al neurotoxicity, and Al is proposed to be involved in the etiology of neurodegenerative diseases. Pb is a widespread environmental hazard, and the neurotoxic effects of Pb are a major public health concern. In spite of the numerous efforts and the accumulating evidence in this area of research, the mechanisms of Al and Pb neurotoxicity are still not completely elucidated. This review will particularly address the involvement of oxidative stress, membrane biophysics alterations, deregulation of cell signaling, and the impairment of neurotransmission as key aspects involved Al and Pb neurotoxicity.